DANIEL KARIMI

STUDENT RESEARCHER (COMPUTER VISION & ML)

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ABOUT ME

Engineering student at **Universidad Politécnica de Madrid** (expected graduation 2025/2026), pursuing a Bachelor's degree with a deep specialization in Computer Science, **Computer Vision, and Machine Learning**. I am deeply passionate about algorithmic foundations and solving complex research problems using advanced Deep Learning techniques.

My academic and practical experience centers around **Artificial Intelligence applied to real-world scenarios**. I have extensive experience working with PyTorch, TensorFlow, and developing custom YOLO architectures for scientific research, including work that is currently pending publication in a peer-reviewed scientific journal.

I am seeking a Student Researcher role where I can apply my background in **Data Science, Machine Learning, and Computer Vision** to push the boundaries of technology alongside world-class researchers at Google.

TECHNICAL STACK

MACHINE LEARNING & AI

Deep Learning • Computer Vision • PyTorch / TensorFlow • YOLO (v5/v8) Architectures • Algorithmic Optimization • Image Processing

DATA SCIENCE

Statistical Modeling • Data Pipelines • Pandas & NumPy • Dataset Curation • Data Normalization • Data Visualization

COMPUTER SCIENCE

Python Applications • Data Structures • Algorithms • C / C++ • Software Engineering • Automation Scripts

ACADEMIC RESEARCH

Scientific Documentation • Literature Review • Experiment Design • Quantitative Analysis • Metrics Evaluation • Peer Collaboration

SYSTEMS & TOOLS

Unix / Linux OS • Git / GitHub • Jupyter Notebooks • Docker • Bash Scripting • Cloud Integration

ENGINEERING FOUNDATIONS

Signal Processing (DSP) • System Architecture • Hardware & Embedded • Microprocessors • Testing & QA • Problem Solving

Languages: Spanish (Native) • English (B2 – Aptis ESOL, British Council)

PROFESSIONAL EXPERIENCE

AI Engineer Intern | CSIC – Institute of Physical and Information Technologies (ITEFI)

Sep 2025 – Feb 2026

- Collaborated as a core Student Researcher with the RESULT scientific group, applying advanced **Computer Vision and Machine Learning** techniques to biological image analysis.
- Engineered end-to-end data processing pipelines to handle massive datasets, focusing on robust curation, normalization, and optimization of raw data for algorithmic training.
- Built, trained, and fine-tuned predictive Deep Learning models using **PyTorch and YOLOv8 architectures**, evaluating performance metrics to optimize inference accuracy.
- Conducted rigorous quantitative analysis and performance evaluations of ML models, directly contributing to a novel methodology pending publication in a peer-reviewed scientific journal.
- Authored comprehensive scientific and technical documentation bridging complex data science concepts with engineering solutions for cross-disciplinary peers.
- Operated predominantly in Python/Linux environments, managing dependencies and optimizing algorithmic foundations for research tasks.

Machine Learning | Computer Vision | PyTorch | Data Science | Scientific Research | Optimization

EDUCATION

Telecommunications Systems Engineering | **Universidad Politécnica de Madrid (ETSIT-UPM)** *Sep 2020 – Jul 2025*

- **Programming & Algorithms:** Data structures, OOP, algorithm design and optimization (C, Python, Java).
- **Networking & Systems:** TCP/IP architecture, network protocols, operating systems, distributed systems.
- **Digital Signal Processing:** Spectral analysis, digital filtering, real-time DSP algorithms.
- **Electronics & Embedded Systems:** Microprocessors, analog/digital circuits, hardware-software integration.

Java | C | Python | Networking | DSP | Algorithms

Technology Baccalaureate, **IES Isabel la Católica** *2018 – 2020*

Science and Technology track. Honours. Strong foundation in mathematics and physics.

CERTIFICATIONS

AI and Productivity (Google): Efficient use of AI tools and automation workflows. •

Generative AI (MIT): Implementation of GenAI solutions and sentiment analysis. •

Internet of Things (MIT): IoT solutions for smart homes and precision agriculture. •

Digital Transformation (MIT): Technology strategy and business digitalization. •

Aptis ESOL B2 (British Council): Official English certification accrediting full linguistic competence (CEFR). Assessment of Reading, Writing, Listening and Speaking, demonstrating ability to work in professional environments and understand complex technical language.

ACADEMIC & RESEARCH PROJECTS

CellTracker (Research Paper) NOVEL ML APPLICATION

PYTORCH • DEEP LEARNING • ALGORITHMS

Algorithmic cell tracking framework. Pending publication in scientific journal.

Academic research project proposing a novel application of custom YOLOv8-based detection models combined with algorithmic tracking (Kalman Filters). Focused on improving predictive accuracy and generating quantitative statistical insights for biological datasets.

- Designed end-to-end Machine Learning pipelines for dataset preprocessing, model training, and performance evaluation.
- Demonstrated strong capabilities in algorithmic optimization and scientific methodology.

MigraCell Accelerator COMPUTER VISION

PYTHON • YOLOV8-SEG • OPTIMIZATION

Deep Learning architecture for automated cell migration analysis.

Developed a scalable Machine Learning solution that drastically reduces experimental inference time (from hours to minutes). Utilized YOLOv8-seg for pixel-perfect segmentation and automated data extraction for large-scale analysis.

- Applied Deep Learning principles to solve complex, real-world analytical bottlenecks.
- Engineered reproducible data structures for generating publication-ready quantitative reports.